TOR INDEPENDENCE FOR A BASE CHANGE SQUARE WITH TWO FLAT LEGS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 20304-20334, immediately following the statement at 20289-20302.

Editorial note. One cross-chapter pointer (More on Algebra) is recorded in assembly.external_references.

Editorial note. Same-name pair note: the spaces side of this label delegates its core case here, so the pair is counted once, here.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 20306 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Double base change of flat algebras. More on Algebra, lemma more-algebra-lemma-double-base-change, cited in the proof. Supplies the algebra identity used to compare the two derived tensor products.
- Flatness giving derived tensor products. used at proof lines 20322-20323. Replaces the ordinary tensor product by the derived one along the flat map A' -> B'.
- Tor independence as vanishing of higher Tor. the definition used in the statement's conclusion. Fixes what has to be shown, namely that the higher Tor groups of the two algebras over the fibre product vanish.

Source reference. Original reference 08YU: focused-seeds/geometry/S9-morph-tor-independence-case-schemes/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} Z' & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X' & \longrightarrow & S' \end{array}$$

Let $S \to S'$ be a morphism. Denote by X and Y the base changes of X' and Y' to S. Assume $Y' \to S'$ and $Z' \to X'$ are flat. Then $X \times_S Y$ and Z' are Tor independent over $X' \times_{S'} Y'$.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The question is local, hence we may assume all schemes are affine (some details omitted). Observe that

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & X' \times_{S'} Y' \\ \downarrow & & \downarrow \\ X & \longrightarrow & X' \end{array}$$

is cartesian with flat vertical arrows. Write $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$, $X' \times_{S'} Y' = \mathrm{Spec}(B')$. Then $X \times_S Y = \mathrm{Spec}(A \otimes_{A'} B')$. Write $Z' = \mathrm{Spec}(C')$. We have to show

$$\mathrm{Tor}_p^{B'}(A \otimes_{A'} B', C') = 0, \quad \text{for } p > 0$$

Since $A' \to B'$ is flat we have $A \otimes_{A'} B' = A \otimes^{\mathbf{L}}_{A'} B'$. Hence

$$(A \otimes_{A'} B') \otimes^{\mathbf{L}}_{B'} C' = (A \otimes^{\mathbf{L}}_{A'} B') \otimes^{\mathbf{L}}_{B'} C' = A \otimes^{\mathbf{L}}_{A'} C' = A \otimes_{A'} C'$$

The second equality by More on Algebra, Lemma 08YU. The last equality because $A' \to C'$ is flat. This proves the lemma. □